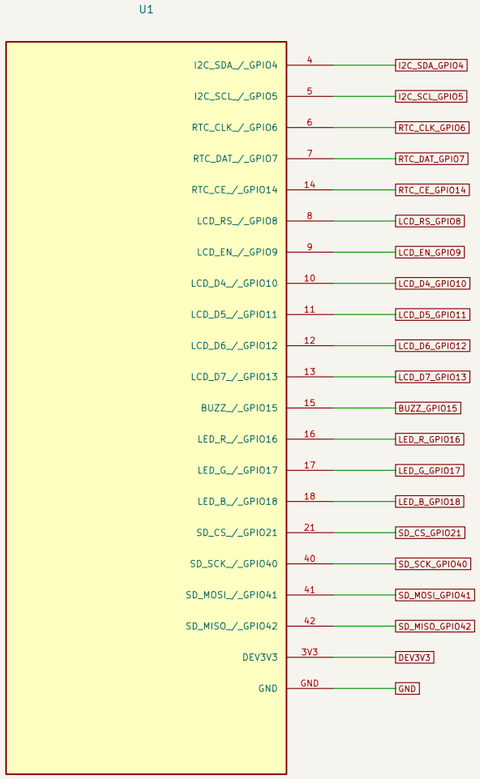
AKENA NEOGUARD | LCD SYSTEM INTEGRATION

Electrical pin map matches the 22 October LCD firmware. Module footprints and power ratings require physical confirmation.

ESP32-S3 controller



Logical GPIO numbers, not header positions.
Connect through J1 on the carrier.
Exact devboard socket is not assumed.

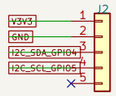
LCD on carrier

File: NeoGuard-lcd.kicad_sch

Power and controller cable

File: NeoGuard-power.kicad_sch

MAX30102 OXIMETER



MLX90614 INFRARED



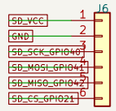
DS1302 RTC



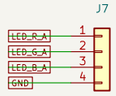
CR2032: trickle charging disabled in firmware.

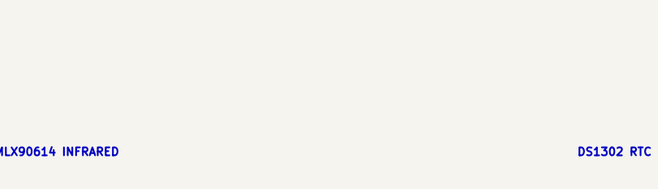
Shared I2C: SDA GPIO4, SCL GPIO5.
Sensor headers are functional interfaces.
Reorder cable wires to match purchased modules.

SD CARD MODULE



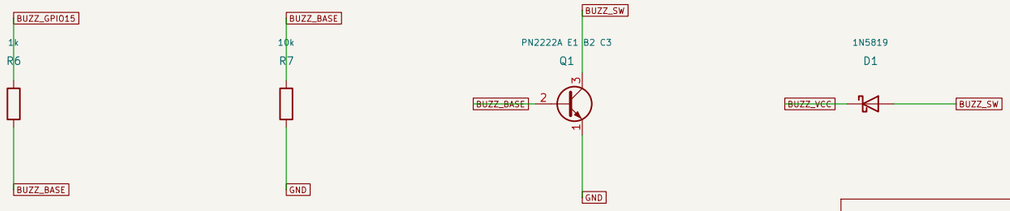
RGB COMMON CATHODE





Each LED colour has its own 330-ohm resistor.

BUZZER DRIVER | GPIO15



PN2222A E1 B2 C3; verify purchased transistor. D1 cathode -> buzzer positive. Driver sized for <=20mA.

LCD pins GPIO8-13 -> sheet 2. SD stays GPIO21/40/41/42.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /
File: NeoGuard.kicad_sch

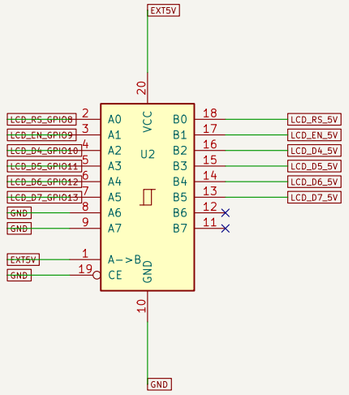
Title: Akena NeoGuard - LCD System

Size: A3	Date: 2026-10-22	**Rev: LCD-C**
KiCad E.D.A. 10.0.4		Id: 1/3

LCD1602 | PHYSICALLY MOUNTED ON THE CARRIER

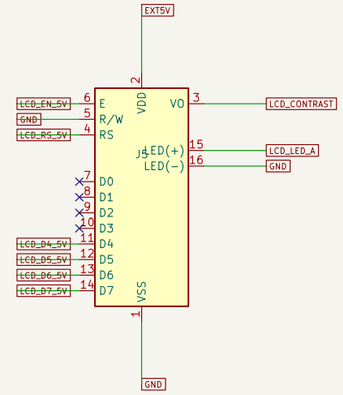
Assumed footprint: WC1602A, 80 x 36 mm, 16 pins at 2.54mm pitch. Verify against the actual display before manufacture.

3.3V –> 5V UNIDIRECTIONAL BUFFER



U2 MUST be SN74AHCT245N.
Symbol represents the hardwired A –> B direction.
Do not substitute a 74HC245.
DIR pin1 = 5V: A –> B.
OE pin19 = GND: outputs enabled.
Unused A6/A7 inputs grounded; B6/B7 left open.

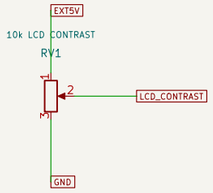
LCD 16 x 2 | 4–BIT MODE



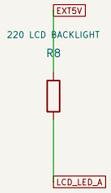
RS=GPIO8 EN=GPIO9
D4=GPIO10 D5=GPIO11
D6=GPIO12 D7=GPIO13

RW grounded: ESP32 never reads a 5V bus.
D0–D3 left unconnected.

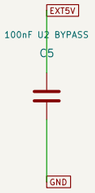
CONTRAST 10k



BACKLIGHT 220R



U2 BYPASS 100nF



LCD BULK 10uF



5V is supplied by regulated EXT5V, independent of ESP32 USB power. All grounds common. Power the controller and carrier together.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /LCD on carrier/
File: NeoGuard–lcd.kicad_sch

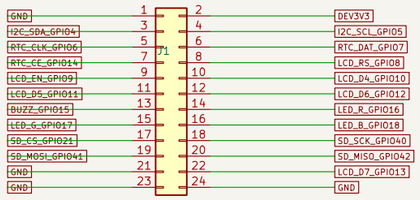
Title: Akena NeoGuard – Mounted LCD and Logic Buffer

Size: A3	Date: 2026–10–22	**Rev: LCD–C**
KiCad E.D.A. 10.0.4		Id: 2/3

POWER SELECTION | CONTROLLER CABLE

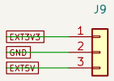
No regulator on this carrier. Connect regulated supplies only. Never short external power to a USB-powered ESP32 rail.

ESP32-S3 FUNCTIONAL CABLE



J1 is not a devboard socket.
Pin22 now carries LCD_D7 / GPIO13.
Map cable wires by GPIO name.
USB remains on the external controller.

REGULATED INPUT



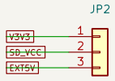
J9: 1=EXT3V3, 2=GND, 3=EXT5V
EXT5V is required for LCD + U2.

3V3 SOURCE



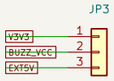
JP1: 1-2 DEV3V3; 2-3 EXT3V3
One shunt only. Do not bridge both.

SD SUPPLY



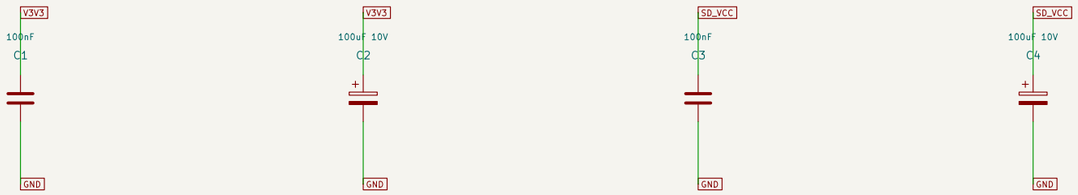
JP2: 1-2 = 3V3; 2-3 = 5V
Select to match the actual SD module.

BUZZER SUPPLY

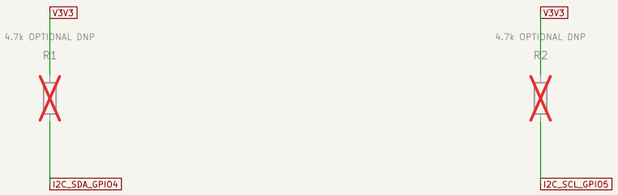


JP3: 1-2 = 3V3; 2-3 = 5V
Select to match the actual buzzer.

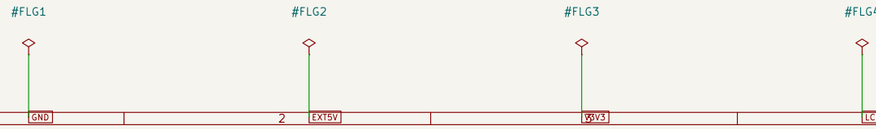
DECOUPLING: SENSOR 3V3 AND SD SUPPLY



OPTIONAL I2C PULL-UPS



R1/R2 DNP when breakout pull-ups already exist. SDA/SCL must be pulled up to 3.3V, never 5V.



GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /Power and controller cable/
File: NeoGuard-power.kicad_sch

Title: Akena NeoGuard – Power and Controller Cable

Size: A3	Date: 2026-10-22	**Rev: LCD-C**
KiCad E.D.A. 10.0.4		Id: 3/3